COVID-19 Guidelines in the Philippines, Singapore, the United Kingdom, and the United States: A Comparative Mixed Method Study

Jasper Kyle Catapang

University of Birmingham, the United Kingdom

Outline

I. Why study COVID-19 guidelines?
II. Analyses
    A. N-grams
        1. Unigrams
        2. Bigrams
    B. Text typology
    C. Textual patterns
    D. Signs
    E. Relational values of grammatical features
    F. Sentiment analysis
    G. Languages involved

Why study COVID-19 guidelines?

- Making information clear, comprehensive, and available, could literally be a matter of life and death.

Why study COVID-19 guidelines?

- Economy and quality of healthcare are factors that cannot be rapidly changed in times of crises, although crafting effective messages in handling such scenarios can be improved relatively faster.

Why study COVID-19 guidelines?

- While we should hold our governments responsible for pandemic responses, there are matters that are left to the citizens. These can be maximized if the guidelines are crafted to optimal persuasive effect.

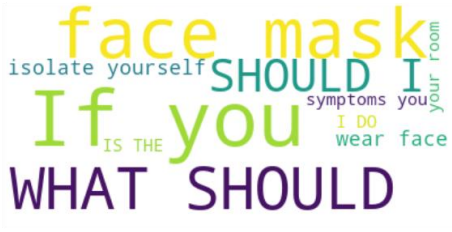
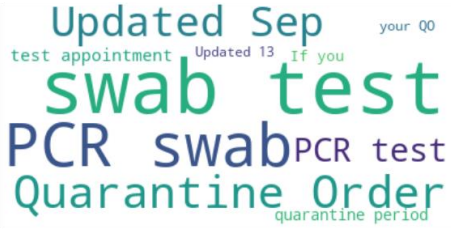
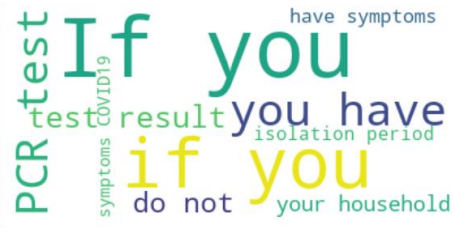
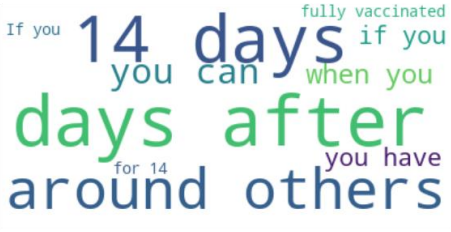
Data

- 4 countries
- 4 types of guidelines
- ~~16 total guidelines~~
- 15 total guidelines

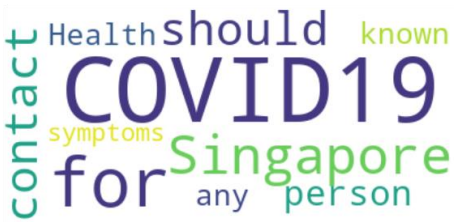
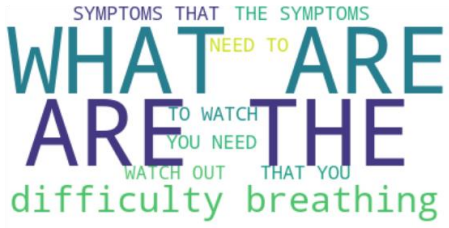
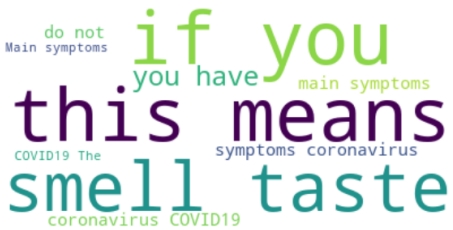
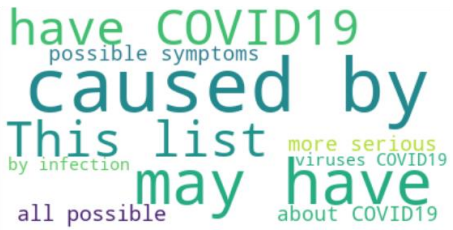
N-grams

- In computational linguistics, an n-gram is commonly a sequence of n words.
- We obtain the n-grams to find individual words or sets of words that occur frequently.
- N-gram frequencies reveal hints about the linguistic properties of a text or speech corpus.

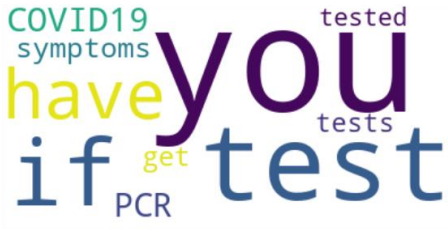
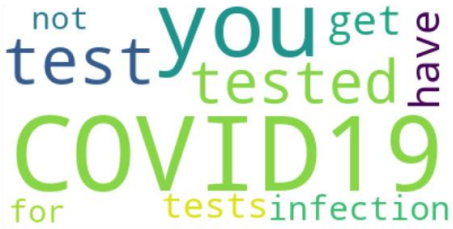
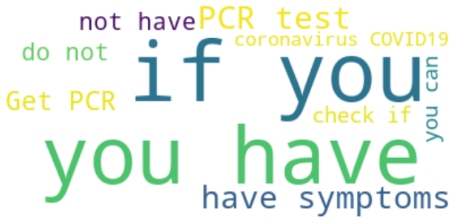
Quarantine guidelines

	Philippines	Singapore	United Kingdom	United States
unigrams				
bigrams				

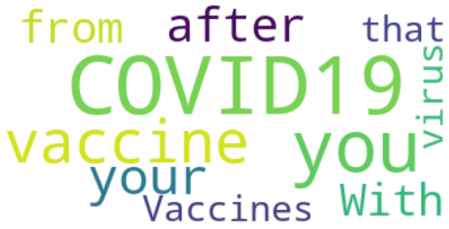
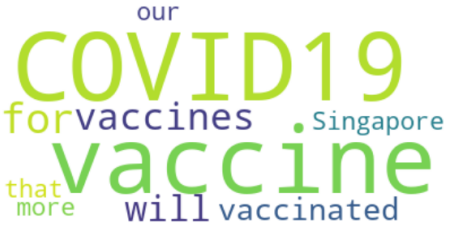
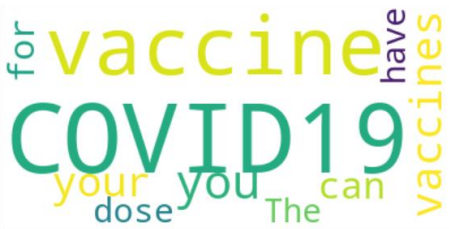
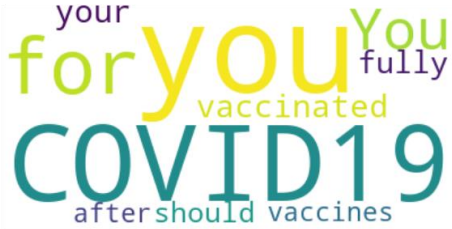
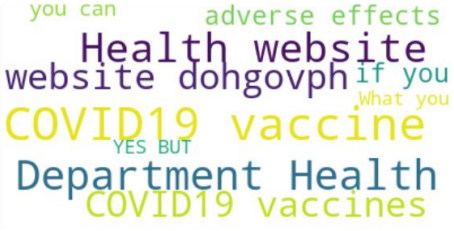
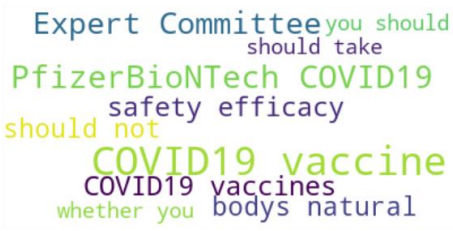
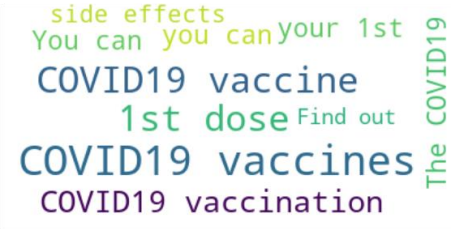
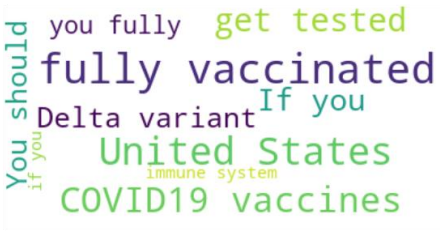
Symptom guidelines

	Philippines	Singapore	United Kingdom	United States
unigrams				
bigrams				

Testing guidelines

	Philippines	Singapore	United Kingdom	United States
unigrams				
bigrams				

Vaccination guidelines

	Philippines	Singapore	United Kingdom	United States
unigrams				
bigrams				

Text typology

This is according to Longacre (1974)’s four type of discourse:

1. Narrative
2. Expository
3. Procedural
4. Behavioural

Text typology

- COVID-19 guidelines are expected to be both procedural and behavioural (persuasive).
- Guidelines, in general, are expected to enumerate steps the recipient should perform or observe (procedural).
- Guidelines are also expected to be persuasive or else the procedure would not be followed or obeyed.

Text typology: quarantine

Country	Procedural	Persuasive	Expository
US	✓	✓	✓
UK	✓	✓	✓
SG	✓		✓
PH	**N/A**	**N/A**	**N/A**

Text typology: symptoms

Country	Procedural	Persuasive	Expository
US	✓		✓
UK	✓		✓
SG	✓		✓
PH	✓		✓

Text typology: testing

Country	Procedural	Persuasive	Expository
US	✓	✓	✓
UK	✓	✓	✓
SG	✓	✓	✓
PH	**N/A**	**N/A**	**N/A**

Text typology: vaccination

Country	Procedural	Persuasive	Expository
US	✓	✓	✓
UK	✓	✓	✓
SG	✓	✓	✓
PH	✓	✓	✓

Textual patterns

- It is important to identify the type of discourse structure in COVID-19 guidelines and advisories since these structures also contribute to how messages are crafted.

Textual patterns

- In the context of COVID-19, guidelines are expected to have a general-specific pattern, policies are expected to have a problem-solution pattern, and fact-checking/mythbusters to have a claim-counterclaim pattern

Textual patterns

- As expected, all guidelines adhere to the general-specific pattern.
- It is interesting how each country differs in utilizing such pattern, through the use of bullets, paragraphs, or a combination of both.
- Overuse of bullets risks oversimplification, while overuse of paragraphs risks disengagement by the recipient(s).

Textual patterns

Country – Information	Form of text
US – symptoms	Bullets & paragraphs
US – testing	Bullets & paragraphs
US – quarantine	Bullets & paragraphs
US – vaccination	Bullets & paragraphs
UK – symptoms	Bullets
UK – testing	Paragraphs & bullets
UK – quarantine	Paragraphs & bullets
UK – vaccination	Bullets & paragraphs
SG – symptoms	Bullets
SG – testing	Bullets
SG – quarantine	Paragraphs & bullets
SG – vaccination	Paragraphs & bullets
PH – symptoms	Bullets & paragraphs
PH – testing	N/A
PH – quarantine	Bullets
PH – vaccination	Bullets

Suggested form of text

Symptoms: bullets

- Symptoms are expected to be easy to enumerate

Testing: bullets and paragraphs

- The outline of the testing procedure should be in bullets and specifics of the procedure can be explained in detailed through the use of paragraphs

Quarantine: bullets and paragraphs

- The outline for the quarantine procedure should follow the same format as the one for testing

Vaccination: paragraphs and bullets

- Vaccination guidelines should be supplied with an in-depth explanation of its efficacy and attempts to minimize hesitancy

Signs

- Since some of the COVID-19 guidelines and advisories make use of infographics, the signs utilised to accompany these texts also need to be analysed.
- COVID-19 guidelines with graphics or multimedia included tend to invite more readers/viewers.

Signs

- Only the testing and vaccination guidelines from the United States made use of images

Relational values of grammatical features

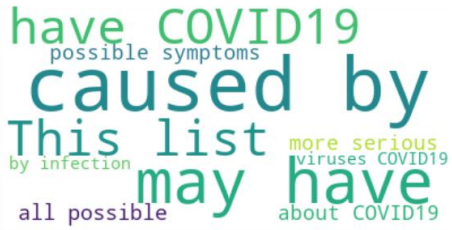
- The use of relational values, borrowed from Fairclough's (1992) framework of critical discourse analysis, aid in the identification of the agenda of the addresser.

Relational values of grammatical features

- Such relational values, like pronouns (you, your, we, etc.), help the addresser persuade the addressee by attempting to situate the addressee in the scenario described.
- Meanwhile, the use of terms like “people”, “persons”, “patients”, dissociate the addressee from the gravity of the pandemic.

Relational values of grammatical features

- Symptom guidelines from the United States and Singapore made no use of relational values of grammatical features.



Sentiment analysis

- All countries utilize neutral tones in their guidelines.
- Guidelines for testing, quarantine, and vaccination carry positive sentiments in the United States.

Sentiment analysis

- The healthy ministry in the United Kingdom uses positive sentiments in their quarantine and vaccination guidelines, encouraging the addressees on the advantages of obeying the two guidelines.
- The UK uses negative tones in its testing guidelines, highlighting the dangers if the addressees are not tested for COVID-19.

Sentiment analysis

- Singapore and the Philippines employ negative tones in their quarantine guidelines and positive tones in their vaccination guidelines.

Languages involved

- The languages that the guidelines are available in entail the intention of inclusivity by the health ministry of a specific country.
- All of the guidelines in this study are available in English.

Languages available

Country – Information	Available in the ff languages:
US – symptoms	English, Spanish, Chinese, Vietnamese, Korean
US – testing	English, Spanish, Chinese, Vietnamese, Korean
US – quarantine	English, Spanish, Chinese, Vietnamese, Korean
US – vaccination	English, Spanish, Chinese, Vietnamese, Korean
UK – symptoms	English
UK – testing	English
UK – quarantine	English
UK – vaccination	English
SG – symptoms	English
SG – testing	English, Chinese, Malay, Tamil
SG – quarantine	English
SG – vaccination	English
PH – symptoms	English, Filipino, Ilokano, Kapampangan, Cebuano, Waray, Maranao, Kinaray-a, Pangasinan, Bikolano, Hiligaynon
PH – testing	N/A
PH – quarantine	English, Filipino, Ilokano, Kapampangan, Cebuano, Waray, Maranao, Kinaray-a, Pangasinan, Bikolano, Hiligaynon
PH – vaccination	English

Summary

- The health ministries in the United States and the Philippines have included the most number of languages their guidelines are available in.
- All countries utilize neutral tones in their guidelines, although the United States makes use of positive tones as well, among other techniques by the United Kingdom (use of negative), and Singapore and Philippines (use of both negative and positive).

Summary

- Pronouns help the addresser persuade the addressee by attempting to situate the addressee in the scenario described.
- Symptom guidelines from the United States and Singapore made no use of pronouns and may cause addressee dissociation from the severity of the COVID-19 pandemic.

Summary

- Only the United States made use of images in their guidelines.
- COVID-19 guidelines are expected to be both procedural and persuasive.
- The health ministry of Singapore crafted quarantine guidelines that lack persuasive power.
- The guidelines have improved upon this expectation by being expositional also.

Summary

CONCLUSION: The COVID-19 guidelines from the four countries are up to standard, and some even exceed the baseline.

COVID-19 Guidelines in the Philippines, Singapore, the United Kingdom, and the United States: A Comparative Mixed Method Study

Jasper Kyle Catapang

University of Birmingham, the United Kingdom